

Appendix A - Complex Numbers

Quadratic formula :

$$x = \frac{-b \pm \sqrt{b^2 + 4ac}}{2a}$$

→ complex numbers
allows $\sqrt{-\#}$

Basics :

define $i = \sqrt{-1}$, so then $i^2 = -1$

$$\rightarrow \mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}$$

$$\Rightarrow \text{addition : } (x + yi) + (a + bi) = (x + a) + (y + b)i$$

$$\Rightarrow \text{multiplication : } (x + yi)(a + bi) = (ax - by) + (ay + bx)i$$

e.g.

$$\text{Compute } (1 + 3i) + (2 - 5i) \text{ \& } (3 + 2i)(1 - i)$$

$$\begin{aligned}(1 + 3i) + (2 - 5i) &= (1 + 2) + (3i - 5i) \\ &= 3 + (-2)i\end{aligned}$$

$$\begin{aligned}(3 + 2i)(1 - i) &= 3 - 3i + 2i - 2i^2 \\ &= 3 - 2(-1) + (-3 + 2)i \\ &= 5 + (-1)i\end{aligned}$$

e.g.

$$\text{compute } (2 + i) + [(1 + i)(2 + 4i)]$$

$$\begin{aligned}&= (2 + i) + [2 + 4i + 2i + 4i^2] \\ &= 2 + i + 2 + 6i + 4(-1)\end{aligned}$$

$$= 4 - 4 + 7i$$

$$= 7i$$

Arithmetic Properties :

let $z = x + yi$, $w = a + bi \in \mathbb{C}$

1. every $\mathbb{R}$ is complex ($t = t + 0i$)

2. Commutative , $zw = wz$

3. Multiplicative , $1z = z$

4. Additive , $z + w = w + z$

5. identity , $z + 0 = z$

6. zero product , $0 \cdot z = 0$

=> behave like real numbers

Reciprocal & Division :

let $z = x + yi \dots$

$$z^{-1} = \frac{1}{x^2 + y^2} (x - yi)$$

→ division given $w \neq 0$: $\frac{z}{w} = z \cdot w^{-1}$

e.g.

find $(1-i)^{-1}$, and compute $\frac{(2+3i)}{1-i}$

$$(1-i)^{-1} = \frac{1}{(1)^2 + (-1)^2} (1-i) = \frac{1}{2} (1-i)$$

$$\begin{aligned}\frac{(2+3i)}{1-i} &= (2+3i)(1-i)^{-1} \\ &= \frac{1}{2} (2+3i)(1-i) = \frac{1}{2} (2-2i+3i-3i^2) \\ &= \frac{1}{2} (5+i)\end{aligned}$$

Geometric Interpretation, Conjugate & length/Modulus :

A complex number $z = x + yi$ is a point on the xy -plane

→ Conjugate, reflection across x -axis :

$$\bar{z} = x - yi$$

→ length/Modulus :

$$|z| = \sqrt{x^2 + y^2}, \text{ from origin}$$

e.g.

let $z = 1 + 3i$, compute $\bar{z}$ & $|z|$

$$\begin{aligned}\bar{z} &= x - yi \\ &= 1 - 3i\end{aligned}$$

$$\begin{aligned}|z| &= \sqrt{(1)^2 + (3)^2} \\ &= \sqrt{10}\end{aligned}$$

Conjugate & Length Properties :

for any $z, w \in \mathbb{C} \dots$

$$1. \overline{z+w} = \overline{z} + \overline{w} \quad , \quad \overline{zw} = \overline{z} \overline{w}$$

$$2. \overline{\left(\frac{z}{w}\right)} = \frac{\overline{z}}{\overline{w}} \quad , \quad \overline{(\overline{z})} = z$$

$$3. z \overline{z} = |z|^2$$

$$4. z^{-1} = \frac{\overline{z}}{|z|^2}$$

$$5. \left| \frac{z}{w} \right| = \frac{|z|}{|w|} \quad , \quad |z+w| \leq |z| + |w|$$

→ triangle inequality

Polar Form :

rewrite $z = x + yi$ as...

$$z = r(\cos \theta + i \sin \theta) = r e^{i\theta}$$

$|z|$

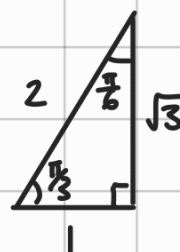
argument of z
 $\arg(z)$

$$\Rightarrow \cos \theta = \frac{x}{r} \quad , \quad \sin \theta = \frac{y}{r}$$

e.g.

express complex number given in polar form by
 $z = 2e^{i\frac{\pi}{6}}$ in $x+yi$ form, then plot z on xy -plane

$$\begin{aligned} z &= 2e^{i\frac{\pi}{6}} \\ &= 2(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}) \\ &= 2\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right) \\ &= \sqrt{3} + i \end{aligned}$$



so the plot point $(\sqrt{3}, 1)$, $w/r = 2$

Multiplication & Powers:

if $z = r e^{i\theta}$ and $w = R e^{i\phi} \dots$

$$zw = rR e^{i(\theta+\phi)}$$

$$z^n = r^n e^{in\theta}$$

e.g.

$z = \sqrt{2} e^{i\frac{2\pi}{3}}$, find z^4 and express as $a+bi$

$$z^4 = r^n e^{i4(\theta)}$$

$$= (\sqrt{2})^4 e^{i4\frac{2\pi}{3}}$$

$$= 4 e^{i\frac{8\pi}{3}}$$

$$= 4 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$$

$$= 4 \left(-\frac{1}{2} + i \frac{\sqrt{3}}{2} \right)$$

$$= -2 + 2\sqrt{3}i$$

$$\frac{8\pi}{3} = 2\pi + \frac{2\pi}{3}$$

Solving Equations, $z^n = a+bi$:

* ① write $z = r e^{i\theta}$

② convert RHS to polar form $R e^{i\phi}$

③ eq'ns: $r^n e^{in\theta} = R e^{i\phi}$

④ compare length: $r^n = R \Rightarrow r = \sqrt[n]{R}$

⑤ compare angles: $n\theta = \phi + 2\pi K$

⑥ $z = \sqrt[n]{R} e^{i(\phi/n + 2\pi K/n)}$, $K = 0, 1, \dots, n-1$

e.g.

find all solutions to eqth $z^3 = 1$

$$z^3 = 1 \quad \Rightarrow \quad r^3 = 1 \Rightarrow r = \sqrt[3]{1} = 1$$

$$r^3 e^{i3\theta} = 1 e^{i0}$$

$$3\theta = 0 + 2\pi K$$

$$\theta = \frac{2\pi}{3} K$$

$$z = 1 e^{i \frac{2\pi}{3} K}, \quad K = 0, 1, 2$$

$$\begin{array}{ccc} \swarrow & \downarrow & \searrow \\ 1 e^{i0} & 1 e^{i \frac{2\pi}{3}} & 1 e^{i \frac{4\pi}{3}} \end{array}$$

$$\begin{aligned} 1 e^0 &= 1 (\cos 0 + i \sin 0) \\ &= 1 (1 + 0) \\ &= 1 \end{aligned}$$

$$\begin{aligned} 1 e^{i \frac{2\pi}{3}} &= 1 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \\ &= \left(-\frac{1}{2} + i \frac{\sqrt{3}}{2} \right) \\ &= -\frac{1}{2} + \frac{\sqrt{3}}{2} i \end{aligned}$$

$$1 e^{i \frac{4\pi}{3}} = \left(-\frac{1}{2} - \frac{\sqrt{3}}{2} i \right)$$

